BPTT: Deriving the Recurrence and the Weight Gradients

From $\partial L/\partial h_t$ to $\partial L/\partial W_{hh}$

Setup: what we want to compute

For a vanilla RNN,

$$a_t = W_{xh}x_t + W_{hh}h_{t-1} + b, \qquad h_t = \phi(a_t), \qquad L = \sum_{k=1}^{T} L_k.$$

Ultimate goal:

$$\frac{\partial L}{\partial W_{hh}}.$$

Why do we first derive a recurrence for the state gradient?

$$\frac{\partial L}{\partial h_t} \quad \text{or} \quad \delta_t := \frac{\partial L}{\partial a_t}$$

Because W_{hh} affects the loss *through the hidden states over time*. Once δ_t is known, the weight gradient becomes simple.

$$\boxed{\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^{T} \delta_t h_{t-1}^{\top}}$$

Step 1: which losses depend on h_t?

The hidden state h_t affects:

- L_t directly,
- L_{t+1} through h_{t+1},
- L_{t+2} through $h_{t+1} \to h_{t+2}$,
- $\ldots$,
- L_T through the whole future chain.

So

$$\frac{\partial L}{\partial h_t} = \frac{\partial}{\partial h_t}(L_t + L_{t+1} + \cdots + L_T).$$

Therefore

$$\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + \frac{\partial}{\partial h_t}(L_{t+1} + \cdots + L_T).$$

Key idea: the derivative is a sum because h_t influences the loss through multiple branches.

Step 2: derive the recurrence for $\partial L / \partial h_t$

All future losses depend on h_t only through h_{t+1}. So by the chain rule,

$$\frac{\partial}{\partial h_t}(L_{t+1} + \cdots + L_T) = \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t}.$$

Hence

$$\boxed{\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t}}$$

Interpretation:

- first term = **current-time contribution**
- second term = **future contribution propagated backward through the recurrence**

Step 3: expand the local transition Jacobian

From

$$a_{t+1} = W_{xh} x_{t+1} + W_{hh} h_t + b, \qquad h_{t+1} = \phi(a_{t+1}),$$
$$\frac{\partial h_{t+1}}{\partial h_t} = \frac{\partial h_{t+1}}{\partial a_{t+1}} \frac{\partial a_{t+1}}{\partial h_t}.$$

Now

$$\frac{\partial a_{t+1}}{\partial h_t} = W_{hh}, \qquad \frac{\partial h_{t+1}}{\partial a_{t+1}} = D_{t+1},$$

where

$$D_{t+1} = \operatorname{diag}\left(\phi'(a_{t+1,1}), \ldots, \phi'(a_{t+1,n})\right).$$

So

$$\boxed{\frac{\partial h_{t+1}}{\partial h_t} = D_{t+1} W_{hh}}.$$

Thus the state-gradient recurrence becomes

$$\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + \frac{\partial L}{\partial h_{t+1}} D_{t+1} W_{hh}.$$

Step 4: define δ_t and get the standard BPTT recursion

Define the pre-activation error

$$\delta_t := \frac{\partial L}{\partial a_t}.$$

Since $h_t = \phi(a_t)$,

$$\delta_t = \frac{\partial L}{\partial h_t} \odot \phi'(a_t).$$

Also, because $a_{t+1} = W_{hh} h_t + \cdots$,

$$\left.\frac{\partial L}{\partial h_t}\right|_{\text{future}} = W_{hh}^\top \delta_{t+1}.$$

So

$$\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + W_{hh}^\top \delta_{t+1}.$$

Multiplying elementwise by $\phi'(a_t)$ gives

$$\boxed{\delta_t = \left(\frac{\partial L_t}{\partial h_t} + W_{hh}^\top \delta_{t+1}\right) \odot \phi'(a_t)}$$

Step 5: now derive the gradient with respect to W_{hh}

Since the same recurrent matrix is used at every time step,

$$\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^{T} \frac{\partial L}{\partial a_t} \frac{\partial a_t}{\partial W_{hh}}.$$

Using $\delta_t = \partial L / \partial a_t$ and

$$a_t = W_{hh} h_{t-1} + \cdots ,$$

we get the local derivative

$$\frac{\partial a_t}{\partial W_{hh}} \rightsquigarrow h_{t-1}.$$

$$\boxed{\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^{T} \delta_t h_{t-1}^{\top}}$$

$$\boxed{\frac{\partial L}{\partial W_{xh}} = \sum_{t=1}^{T} \delta_t x_t^{\top}}, \qquad \boxed{\frac{\partial L}{\partial b} = \sum_{t=1}^{T} \delta_t}.$$

3-step unrolled RNN: direct expansion of $\partial L/\partial W_{hh}$

Let $L = L_1 + L_2 + L_3$. Since W_{hh} is used at all three steps,

$$\frac{\partial L}{\partial W_{hh}} = \frac{\partial L}{\partial a_1}\frac{\partial a_1}{\partial W_{hh}} + \frac{\partial L}{\partial a_2}\frac{\partial a_2}{\partial W_{hh}} + \frac{\partial L}{\partial a_3}\frac{\partial a_3}{\partial W_{hh}}.$$

Defining $\delta_t := \partial L/\partial a_t$ gives

$$\frac{\partial L}{\partial W_{hh}} = \delta_1 \frac{\partial a_1}{\partial W_{hh}} + \delta_2 \frac{\partial a_2}{\partial W_{hh}} + \delta_3 \frac{\partial a_3}{\partial W_{hh}}.$$

Because $a_t = W_{hh} h_{t-1} + \cdots$,

$$\boxed{\frac{\partial L}{\partial W_{hh}} = \delta_1 h_0^\top + \delta_2 h_1^\top + \delta_3 h_2^\top}$$

So the remaining task is: compute $\delta_1, \delta_2, \delta_3$ efficiently.

3-step example: backward recursion for $\delta_3, \delta_2, \delta_1$

At the final step,

$$\delta_3 = \frac{\partial L_3}{\partial h_3} \odot \phi'(a_3).$$

At time 2, a_2 affects both L_2 and the future through a_3:

$$\delta_2 = \left(\frac{\partial L_2}{\partial h_2} + W_{hh}^\top \delta_3 \right) \odot \phi'(a_2).$$

At time 1, the whole future is summarized by δ_2:

$$\delta_1 = \left(\frac{\partial L_1}{\partial h_1} + W_{hh}^\top \delta_2 \right) \odot \phi'(a_1).$$

So in general,

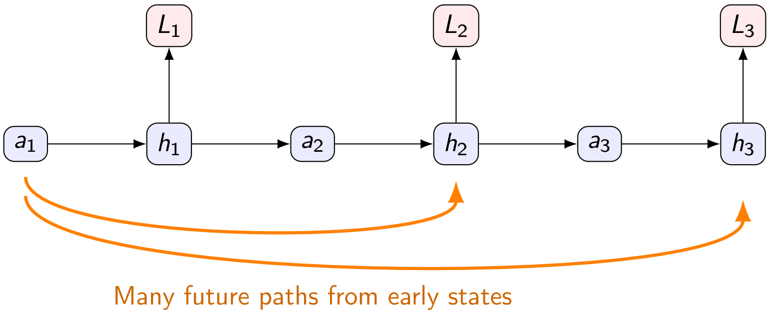
$$\boxed{\delta_t = \left(\frac{\partial L_t}{\partial h_t} + W_{hh}^\top \delta_{t+1} \right) \odot \phi'(a_t)}$$

This recurrence compresses all future paths into a single backward signal.

Why the recurrence matters

Without the recurrence, the derivative at early time steps explodes into many explicit future paths:

$$a_1 \to h_1 \to a_2 \to h_2 \to L_2, \qquad a_1 \to h_1 \to a_2 \to h_2 \to a_3 \to h_3 \to L_3, \quad \dots$$



The backward recursion *summarizes* these paths compactly in $W_{hh}^\top \delta_{t+1}$.

Takeaway

1. First derive the state-gradient recurrence:

$$\frac{\partial L}{\partial h_t} = \frac{\partial L_t}{\partial h_t} + \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t}.$$

2. Convert it to the standard pre-activation recursion:

$$\delta_t = \left(\frac{\partial L_t}{\partial h_t} + W_{hh}^{\top} \delta_{t+1} \right) \odot \phi'(a_t).$$

3. Then the recurrent weight gradient is easy:

$$\frac{\partial L}{\partial W_{hh}} = \sum_{t=1}^{T} \delta_t h_{t-1}^{\top}.$$